Building a Page: A Comprehensive Guide

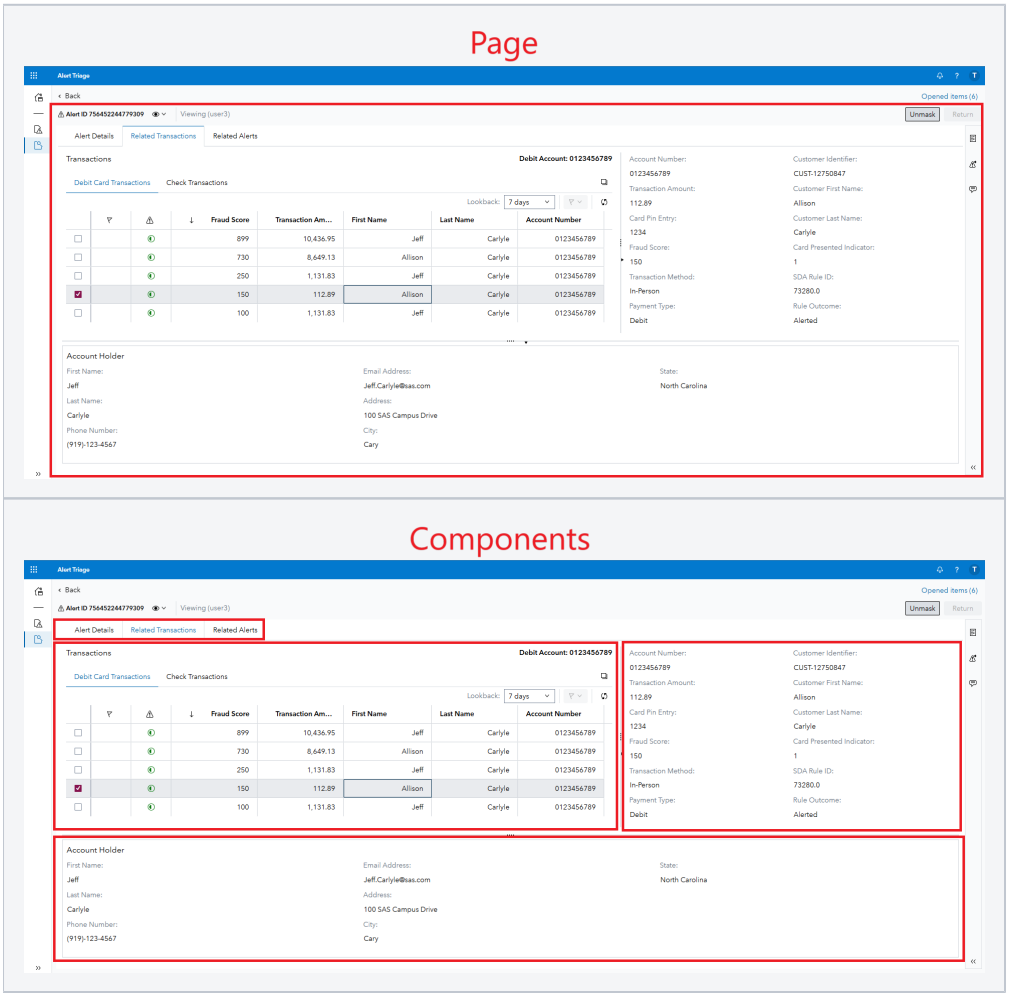
This page serves to break down the different components that comprise pages in SAS Alert Triage and provide examples and tips on how to configure them.

- Pages vs. Components
- Page Definitions
- Components
  - Container Components
  - Content Components
    - Datasources, Landmarks, and Actions
  - Reusable Components
- Example Configuration Files

Expand/ Collapse All Sections

Pages vs. Components

Before diving into *how* to create a page, it's important to understand *what* a page is. Pages allow administrators to customize how alert information is surfaced to end users in SAS Alert Triage. Pages are accessed either through the "Find Alerts" feature or through "Alert Servicing". As of the 2024.09 cadence release, pages are comprised of **components**, allowing page layouts to be highly customizable.



Page Definitions

Now that you have a better grasp on what pages are and how they are differentiated from components, let's create a page definition skeleton to start building a page.

Page definitions require a few variables to be defined:

Variable	Data Type	Description
`id`	string	Unique identifier for the page definition
`entityTypeID`	string	ID of the Entity Type associated with the page definition
`triageTypeID`	string	ID of the Triage Type the page will surface for
`definitionData`	JSON object	Object containing the page's components and the version of the page definition schema. **All pages now use "version": 2**
`optLockVersion`	integer	The page's optLockVersion. -1 updates the previous page configuration.

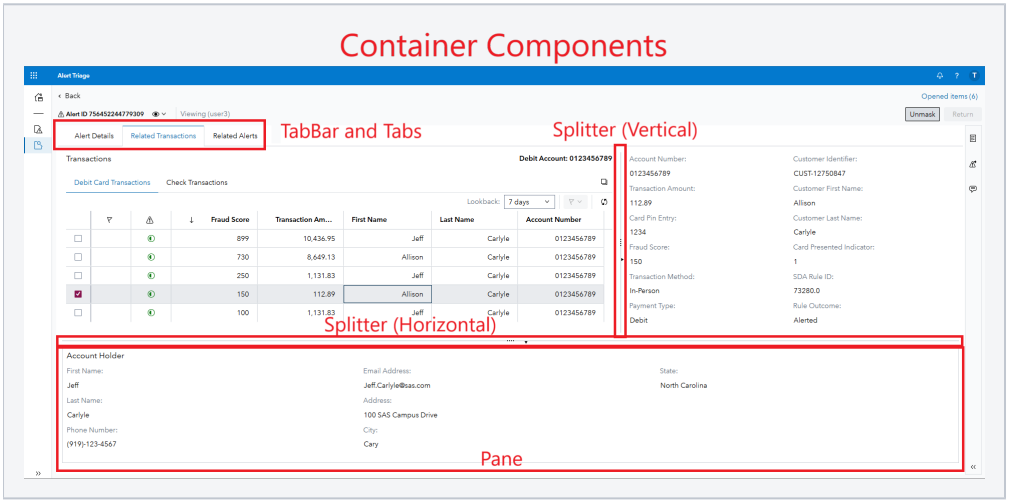
Below, you'll find a template JSON to start building your page.

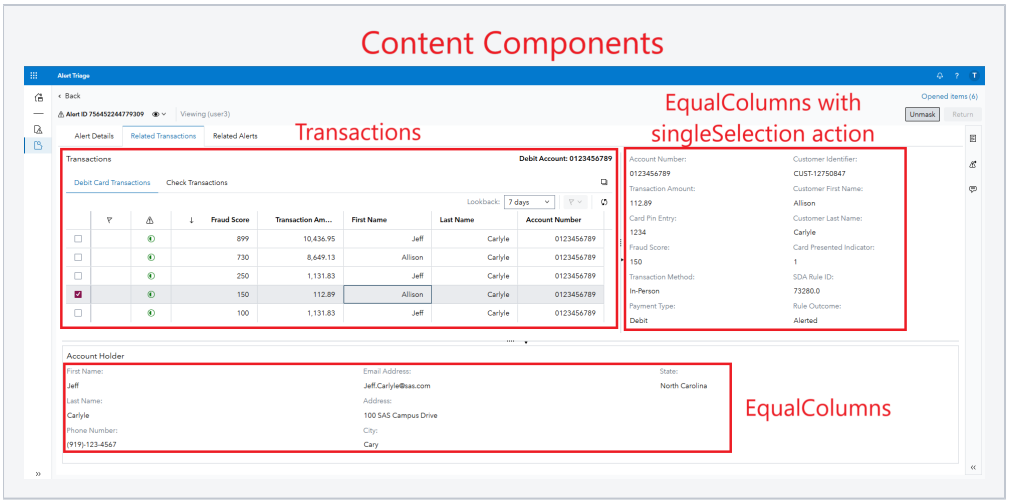
Page Definition Skeleton

```
{
  "id": "example_page_definition",
  "entityTypeID": "entity_type_id",
  "triageTypeID": "triage_type_id",
  "definitionData": {
    // Components go here,
    "version": 2
  },
  "optLockVersion": -1
}
```

Components

Moving on to components, every component can be thought of as either a **container component** or a **content component** depending on whether the component is housing other components or surfacing alert information, respectively.





Additionally, components can either be defined **in-line** (via the "/alerts/pageDefinitions" endpoint) or as **reusable components** (via the "/alerts/componentDefinitions" endpoint and then referenced by the "configID" field in the page definition).

The following variables are used when defining components:

Variable	Data Type	Required?	Description
`id`	string	Yes, if "configID" is not being used	The unique identifier for the component
`type`	string	Yes	The type of component being defined. Container component values: • `"TabBar"` • `"Tab"` • `"Pane"` • `"Splitter"` Content component values: • `"EqualColumns"` • `"Table"` • `"RelatedAlerts"` • `"Transactions"`
`config`	JSON object	Yes, if "configID" is not being used	Used to configure component fields
`configID`	string	Yes, if "config" is not being used	Used to reference reusable components, RelatedAlerts grid definitions, and Transaction Viewer definitions
`datasources`	Array of JSON objects	Depends on the component	Surfaces data to be used within the component. More on this later. Components that support datasources: • `"TabBar"` • `"Pane"` • `"Splitter"` • `"EqualColumns"` • `"Table"` • `"RelatedAlerts"` • `"Transactions"`

For now, let's focus on components defined within the page definition (in-line components). We will discuss reusable components later in this guide.

Page with Component Skeleton

```
{
  "id": "example_page_definition",
  "entityTypeID": "entity_type_id",
  "triageTypeID": "triage_type_id",
  "definitionData": {
    "id": "current_alert_info", // Component unique identifier
    "type": "EqualColumns", // Component type
    "datasources": [
      // Component datasources go here
    ],
    "config": {
      // Component configuration variables go here
    },
    "version": 2
  },
  "optLockVersion": -1
}
```

Container Components

Now that we have a page skeleton set up, let's explore the different **container components** that can go in a page definition.

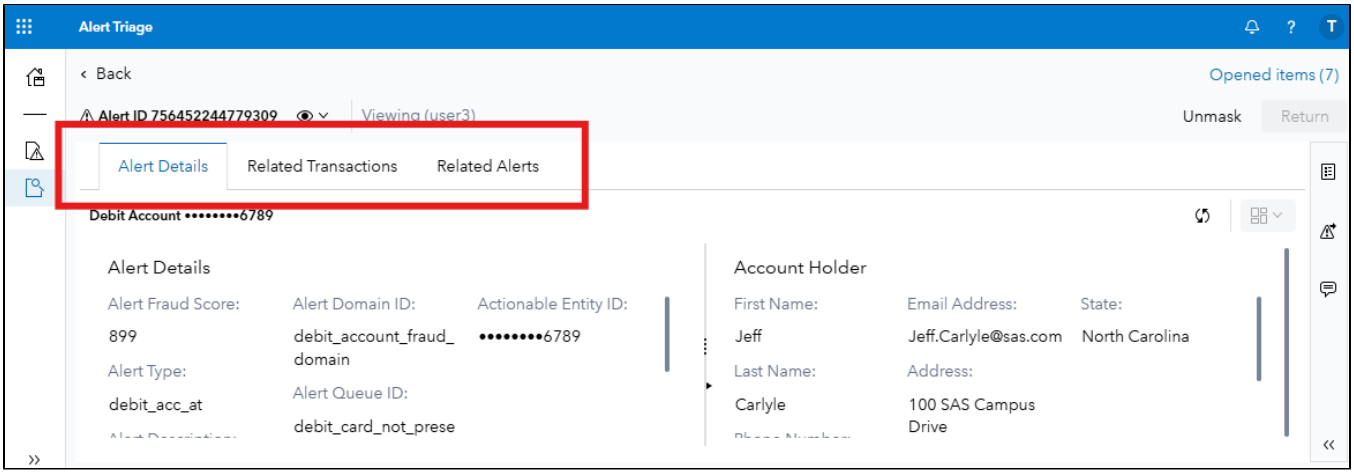
Since container components house other components, every container component must have at least one child component. Because of this, the following variables are used by all container components:

Variable	Data Type	Required?	Description
`childComponents`	Array of component JSON objects	Yes	Array of the container component's children
`childComponents[n].defaultSize`	string	No	Component's default height (if page orientation is vertical) or width (if page orientation is horizontal). Accepts value as percent ('%'), pixel ('px'), or root em ('rem'). All sibling components must share the same units.

Container Component Skeleton

```
{
  "id": "example_pane_component",
  "type": "Pane", // Pane is a container component
  "config": {
    "defaultLabel": "Example Pane"
  },
  "childComponents": [
    {
      // Child component information goes here
      "defaultSize": "75%"
    }
  ]
}
```

Now let's take a look at the different container components and their specific variables.



TabBar: groups multiple Tab components together

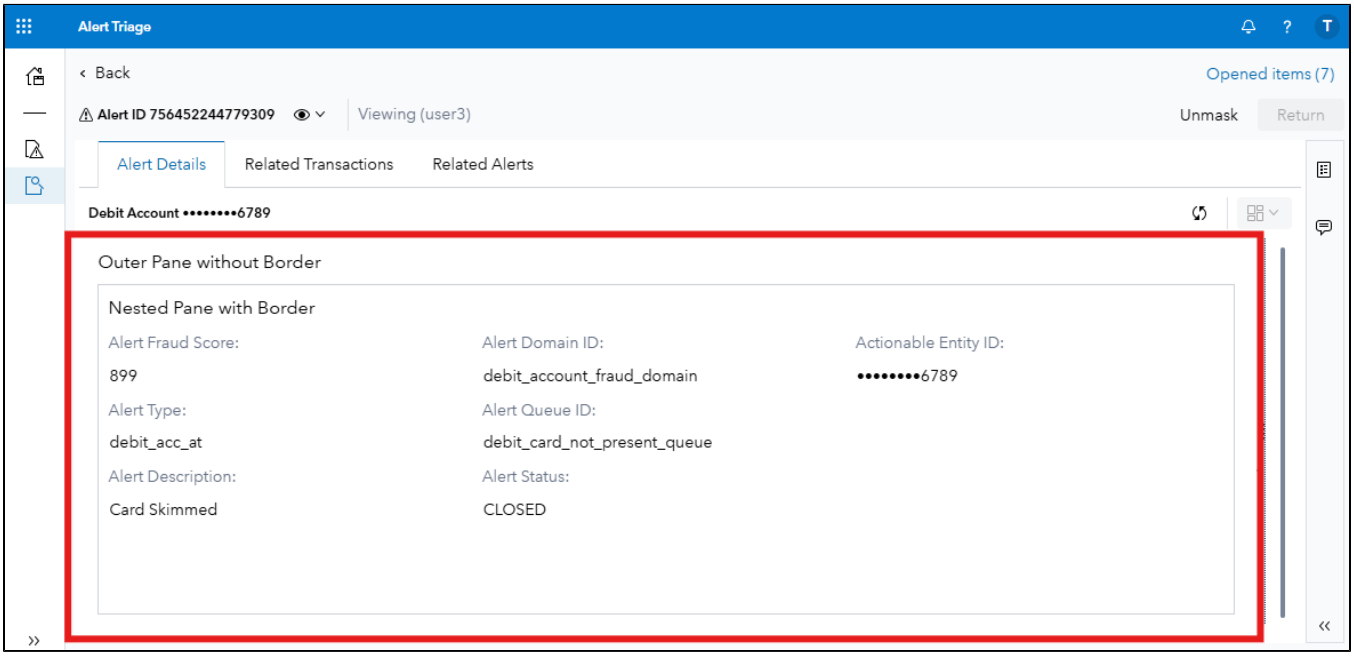
Tab: separates page components into different tabs

Component	'config' Variable	Data Type	Required?	Default Value	Description
TabBar	`compact`	boolean	No	`false`	Compacts Tab size within the TabBar
Tab	`defaultLabel`	string	Yes		Label that appears on the Tab
	`showToolbar`	boolean	No	`false`	Indicates if the toolbar will be shown at the top of the Tab
	`flow`	string	No	`"vertical"`	Indicates the alignment of components within the Tab Acceptable values: • `"vertical"` • `"horizontal"`

Below is an example TabBar component that contains two Tab components

TabBar and Tabs JSON Skeleton

```
{
  "id": "tab_bar_component",
  "type": "TabBar",
  "config": { "compact": false},
  "childComponents": [
    {
      "id": "tab_1",
      "type": "Tab",
      "config": {
        "defaultLabel": "Tab 1",
        "showToolbar": true,
                "flow": "vertical"
      },
      "childComponents": [
        {
          // Components that go here will surface under Tab 1
        }
      ]
    },
    {
      "id": "tab_2",
      "type": "Tab",
      "config": {
        "defaultLabel": "Tab 2",
        "showToolbar": false,
                "flow": "horizontal"
      },
      "childComponents": [
        {
          // Components that go here will surface under Tab 2
        }
      ]
    }
  ]
}
```



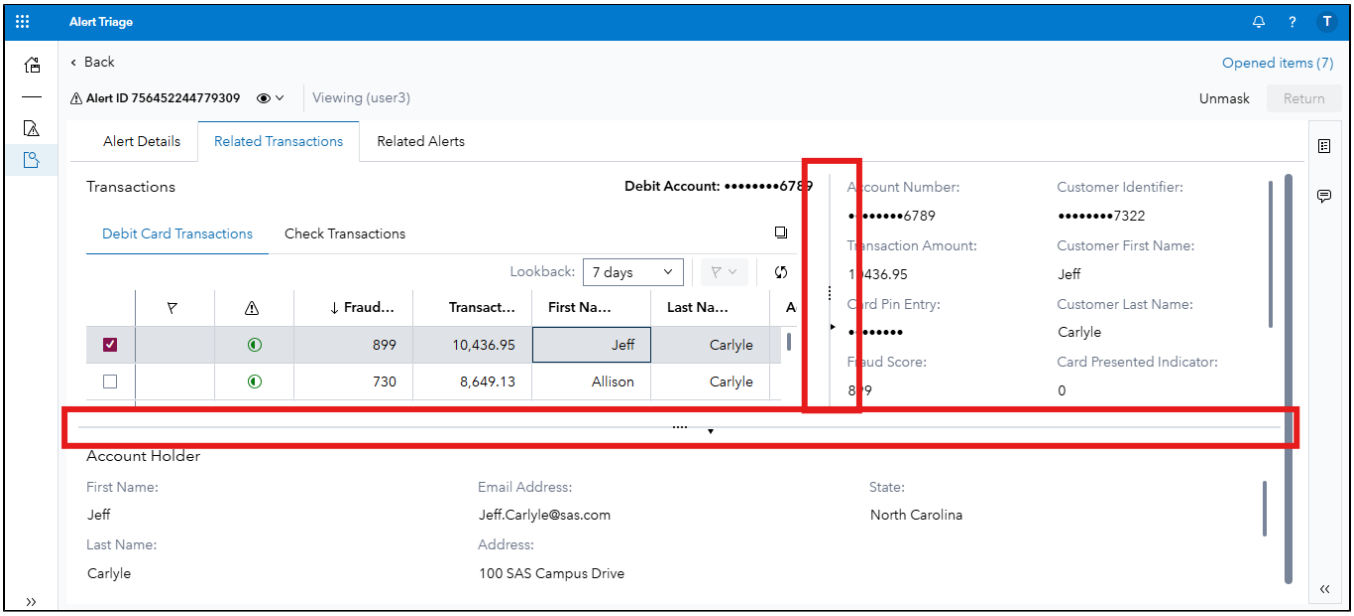
Pane: contains and nests components

'config' Variable	Data Type	Required?	Default Value	Description
`defaultLabel`	string	No		Label that appears at the top of the Pane
`border`	boolean	No	`false`	Indicates if the Pane will have a border around it
`showToolbar`	boolean	No	`false`	Indicates if the toolbar will appear at the top of the Pane
`collapsible`	boolean	No	`false`	Indicates if the Pane is collapsible
`spacing`	string	No	`"xsmall"`	Determines the gap size between the Pane's child components Acceptable values: • `"xsmall"` • `"small"` • `"medium"` • `"large"`
`flow`	string	No	`"vertical"`	Indicates the alignment of the Pane's child components Acceptable values: • `"vertical"` • `"horizontal"`

Below is an example Pane component that contains a nested Pane within it.

Pane JSON Skeleton

```
{
  "id": "outer_pane_component",
  "type": "Pane",
  "config": {
    "defaultLabel": "Outer Pane without Border",
    "border": false,
    "showToolbar": false,
    "collapsible": false,
    "spacing": "xsmall",
    "flow": "vertical"
  },
  "childComponents": [
    // Components that go here will be surfaced in the outer_pane_component
    {
      "id": "nested_pane_component",
      "type": "Pane",
      "config": {
        "defaultLabel": "Nested Pane with Border",
        "border": true
      },
      "childComponents": [
        {
          // Components that go here will be surfaced in the nested_pane_component
        }
      ]
    }
  ]
}
```



Splitter: Divides the page into two resizable sections, either vertically or horizontally. Each section contains a child component(s).

'config' Variable	Data Type	Required?	Default Value	Description
`showToolbar`	boolean	No	`false`	Indicates whether or not the toolbar is shown at the top of the splitter.
`splitbarType`	string	No	`"horizontalCollapseUp"`	Determines how the page is split. Horizontally split pages have 2 rows and 3 columns (2x3), while vertically split pages have 3 rows and 2 columns (3x2). Accepted values: • `"horizontalCollapseDown"` • `"horizontalCollapseUp"` • `"verticalCollapseLeft"` • `"verticalCollapseRight"` • `"none" (becomes CSS grid of equal rows/ columns based on number of child components)`
`defaultSizeSection`	string	No	`"50%"`	Specifies size of the section that the Splitter collapses on. Accepted values: • 0% through 100%

The Splitter component also has an additional variable for its child components.

Variable	Data Type	Required?	Description
`childComponents[n].coordinates`	Array, [number, number]	Yes	[row, column] of where to place the Splitter's child components Accepted values: • Vertical splitter accepts coordinate values from [0, 0] to [2, 1] • Horizontal splitter accepts coordinate values from [0, 0] to [1, 2]

Below is an example Splitter with two reusable child components.

Splitter JSON Skeleton

```
{
  "id": "alert_details_splitter",
  "type": "Splitter",
  "config": {
    "splitbarType": "verticalCollapseRight",
    "defaultSizeSection": "25%",
    "showToolbar": false
  },
  "childComponents": [
    {
      "configID": "alert_details_reusable_component",
      "coordinates": [0, 0]
    },
    {
      "configID": "contact_data_reusable_component",
      "coordinates": [0, 1]
    }
  ]
}
```

Content Components

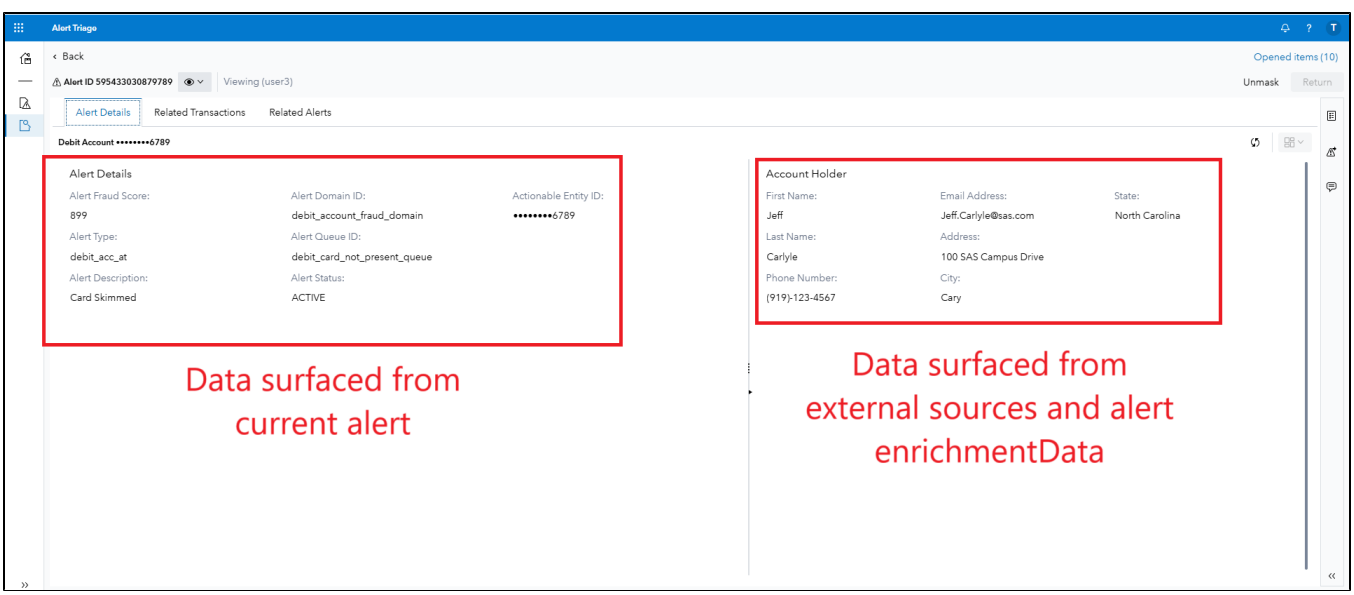
Before moving on to the specific content components, there are a few important concepts to understand: **datasources**, **landmarks**, and **actions**.

Datasources, Landmarks, and Actions

Datasources specify what data is being surfaced in a component and where that data is originating.

Data can be retrieved from:

- The current alert
- Other components on the page
- Data within the SAS ecosystem
- External data sources (retrieved via BOSS)



Datasources are defined by the following properties:

Variable	Data Type	Required?	Default Value	Description
`id`	string	Yes		Unique identifier for the datasource

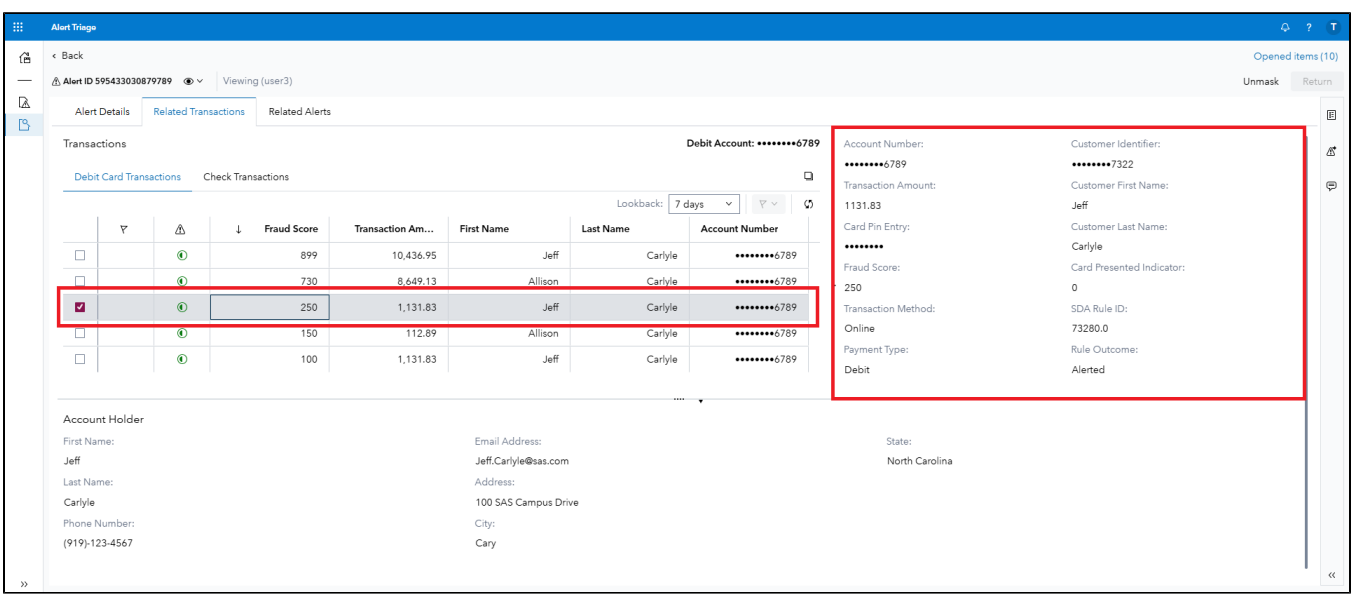
`externalUri`	string	Yes, if no 'request' or 'path' is defined		Indicates data is fetched externally using BOSS
`request`	JSON object	Yes, if no 'externalUri' or 'path' is defined		Indicates data is fetched from the SAS ecosystem through HTTP
`request.uri`	string	Yes, for 'request'		The endpoint called to fetch data
`request.verb`	string	No	`"POST"`	Verb used in HTTP call Accepted values: • `"GET"` • `"POST"`

`request.headers`	JSON object	No	`"Content-Type": "application/json"`
			
			`"Accept-Type": "application/json"`
				Object containing the HTTP headers

`request.body`	JSON object	No		Additional HTTP parameters to be passed in the body of the POST request
`path`	string	Yes, if no 'externalUri' or 'request' is defined		Indicates data is fetched from alert data. See below table for additional details.

Additionally, namespace **landmarks** are used as substitution variables that indicate where the data is coming from.

Landmark	Description	Datasources that Support this Landmark	Syntax
`@currentAlert`	Signifies data is coming from the current alert	• externalUri • path • request.uri • request.body	• `"{$@currentAlert}"` • `"{$@currentAlert}.enrichmentData"`
`@component`	Signifies data is coming from another component on the page	• externalUri • request.uri • request.body	• `"${@component.<componentID>.<action>.<property>}"`
`@bulkData.<datasourceID>`	Signifies data is coming from another component's datasource	• path	• `"${@bulkData.<bulk_datasource_id>.<optional_subpath_into_data>}"`



Actions can also surface information in response to a user's interaction with the page. In the screenshot above, the EqualColumns component on the right uses the 'path' datasource with a @component landmark and the singleSelection action to display details about the transaction the user selects.

The **@component** landmark is currently the only landmark that supports the use of **actions**.

Action	Landmark	Description
`singleSelection`	`@component`	Used to surface additional information upon end user selections Components that can send information: • Table • Transactions • RelatedAlerts Components that can receive and surface information: • EqualColumns • Table

Below, you'll find a few examples of datasources:

External Data and Enrichment Data Example

```
{
  "id": "contact_data_reusable_component",
  "definitionData": {
    "type": "Pane",
    "config": { "defaultLabel": "Account Holder", "border": false },
    "childComponents": [
      {
        "id": "contact_information_equal_columns",
        "type": "EqualColumns",
        "datasources": [ // Datasources go here
          {
            "id": "external_contact_data",
            "externalUri": "/contact/data/${contactEntityID}" // Data from external source, coming from BOSS
          },
          {
            "id": "enrichment_data",
            "path": "${@currentAlert}.enrichmentData" // Data from alert's enrichment data
          }
        ],
        "config": {
          "fields": [
            {
              "defaultLabel": "First Name",
              "field": "enrichment_data.customer_first_name",
              "type": "string"
            },
            {
              "defaultLabel": "Last Name",
              "field": "enrichment_data.customer_last_name",
              "type": "string"
            },
            {
              "defaultLabel": "Phone Number",
              "field": "external_contact_data.phone",
              "type": "string"
            },
            {
              "defaultLabel": "Email Address",
              "field": "external_contact_data.email",
              "type": "string"
            }
          ]
        }
      }
    ],
    "version": 2
  },
  "optLockVersion": -1
}
```

RelatedAlerts with singleSelection Example

```
{
  "id": "alerts_splitter",
  "type": "Splitter",
  "config": { "splitbarType": "verticalCollapseRight" },
  "childComponents": [
    {
      "type": "RelatedAlerts",
      "configID": "debit_card_fraud_triage_type",
      "coordinates": [0, 0]
    },
    {
      "id": "alerts_action_component",
      "type": "EqualColumns",
      "coordinates": [0, 1],
      "datasources": [ // Datasources go here
        {
          "id": "alert_details",
          "request": {
            "headers": {
              "Accept": "application/vnd.sas.triage.alert+json",
              "Content-Type": "application/vnd.sas.triage.alert+json"
            },
// Signifies singleSelection for RelatedAlerts table
            "uri": "/alerts/alerts/${@component.debit_card_fraud_triage_type.singleSelection.id}",
            "verb": "GET"
          }
        }
      ],
      "config": {
        "fields": [
          {
            "defaultLabel": "Name on Card",
            "field": "alert_details.actionableEntityLabel",
            "type": "string"
          },
          {
            "defaultLabel": "Alert Fraud Score",
            "field": "alert_details.score",
            "type": "number"
          },
          {
            "defaultLabel": "Alert Type",
            "field": "alert_details.alertType",
            "type": "string"
          },
          {
            "defaultLabel": "Alert Description",
            "field": "alert_details.alertTriggerText",
            "type": "string"
          }
        ],
        "numCols": 2,
        "defaultMsgSingleSelection": "Please select an alert to view its details here" // Message that appears
when 0 or 1+ alerts are selected
      }
    }
  ]
}
```

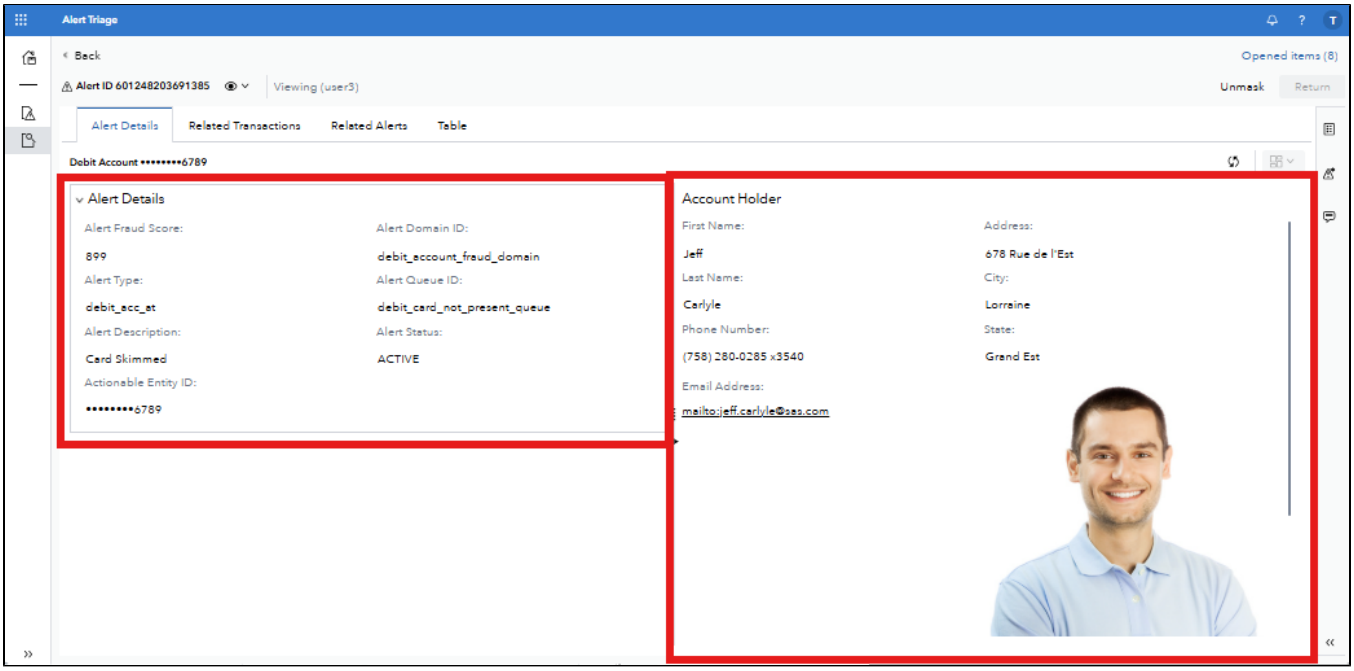
Transactions with singleSelection Example

```
{
  "id": "transaction_splitter",
  "type": "Splitter",
  "config": { "splitbarType": "verticalCollapseRight" },
  "childComponents": [
    {
      "type": "Transactions",
      "configID": "debit_account_transaction_viewer",
      "coordinates": [0, 0]
    },
    {
      "id": "transaction_action_component",
      "type": "EqualColumns",
      "coordinates": [0, 1],
      "datasources": [ // Datasources go here
        {
          "id": "transaction_details",
          "request": {
            "headers": {
              "Accept": "application/vnd.sas.triage.transaction+json",
              "Content-Type": "application/vnd.sas.triage.transaction.details.query+json"
            },
            // URI for obtaining transaction data requires transactionTypeID
            "uri": "/alertsTransaction/transactions/${@component.debit_account_transaction_viewer.
singleSelection.transactionID}?transactionTypeID=${@component.debit_account_transaction_viewer.singleSelection.
transactionTypeID}",
            "verb": "GET"
          }
        }
      ],
      "config": {
        "fields": [
          {
            "defaultLabel": "Account Number",
            "field": "transaction_details.transactionMessage.message.debitaccount.number",
            "type": "string"
          },
          {
            "defaultLabel": "Card Presented Indicator",
            "field": "transaction_details.transactionMessage.message.cardfinancial.cardPresentedInd",
            "type": "string",
            "referenceListID": "card-present-indicator"
          },
          {
            "defaultLabel": "SDA Rule ID",
            "field": "transaction_details.transactionMessage.message.sas.rre.rules[0].refId",
            "type": "string"
          },
          {
            "defaultLabel": "Rule Outcome",
            "field": "transaction_details.transactionMessage.message.sas.decision.outcomeName",
            "type": "string"
          }
        ],
        "numCols": 2,
        // Message will appear if 0 or 1+ transactions is selection in the Transaction Viewer
        "defaultMsgSingleSelection": "Please select a transaction to view its details here"
      }
    }
  ]
}
```

bulkData Example

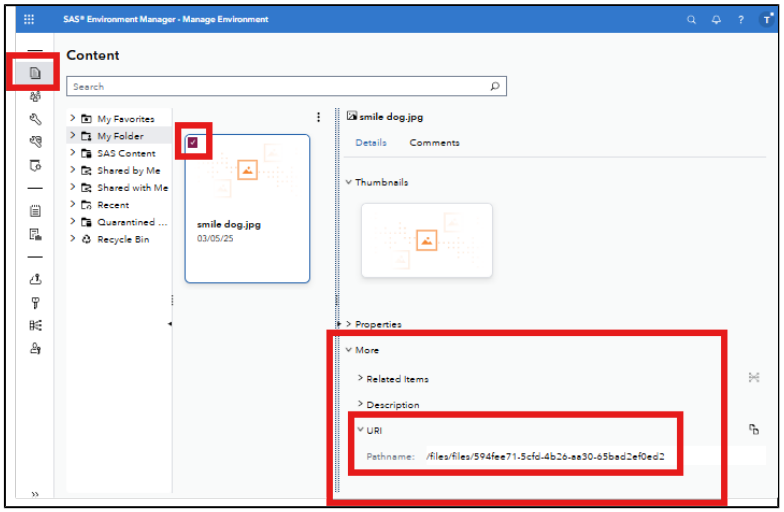
```
{
  "id": "example_pane_with_datasource",
  "type": "Pane",
  "datasources": [ // Container components can have datasources too!
    {
      "id": "contact_data",
      "externalUri": "/contact/data/${contactEntityID}"
    }
  ],
  "config": { "defaultLabel": "Pane" },
  "childComponents": [
    {
      "id": "example_equal_columns",
      "type": "EqualColumns",
      "datasources": [
        {
          "id": "pane_bulk_data", // Content components always have datasources
          "path": "${@bulkData.contact_data}" // Uses the id of the Pane datasource to fetch data
        }
      ],
      "config": {
        "fields": [
          {
            "defaultLabel": "Customer Name",
            "field": "pane_bulk_data.name", // Points to EqualColumns datasource definition, which uses
@bulkData landmark
            "type": "string"
          },
          {
            "defaultLabel": "Customer DOB",
            "field": "pane_bulk_data.dob",
            "type": "datetime"
          },
          {
            "defaultLabel": "Customer SSN",
            "field": "pane_bulk_data.ssn",
            "type": "string"
          },
          {
            "defaultLabel": "Customer Phone Number",
            "field": "pane_bulk_data.phone",
            "type": "string"
          }
        ]
      }
    }
  ]
}
```

Now that we have covered how to leverage **content components**, let's move on to the different types of content components and how to configure them.



EqualColumns: surfaces key/ value information in equal-width columns

'config' Variables	Data Type	Required?	Default Value	Description
`fields`	Array of JSON objects	Yes		Populates the fields in the EqualColumns component
`fields [n]. defaultLabel`	string	No		Label that will be rendered for the field
`fields [n].field`	string	Yes		Path to the field value
`fields [n].type`	string	No	`"string"`	Data type of the field Accepted values: • `"string"` • `"boolean"` • `"number"` • `"date"` • `"datetime"` • `"image"` • `"hyperlink"`

`fields[n].imgSrc`	string	Yes if fields [n].type is "image"		Image path. Images must be contained within the SAS ecosystem (i.e. "Content" tab within Environment Manager). Surfacing this image can be done in several ways, such as leveraging BOSS to pull the image from the file service's database. Another way is directly referencing the file path with an authenticated user (shown below). /files API Documentation: • `{scheme}://{host}/files/files/{fileId}/content` The file ID can be found by clicking the image, expanding "More", and then expanding "URI":  For images to appear within the page, ensure the file is shared with all necessary users. If users do not have access to the image, it will not appear on the page.
`numCols`	number	No	2	Desired number of equal-width columns
`defaultMsgSingleSelection`	string	No	`"Make a single selection to fetch data for this component."`	Message displayed if datasource uses singleSelection action

Below is an example EqualColumns component JSON

EqualColumns Example JSON

```
{
  "id": "example_equal_columns",
  "type": "EqualColumns",
  "datasources": [
    // Datasources go here
    {
      "id": "bulk_contact_data",
      "externalUri": "/contact/data/${contactEntityID}"
    },
    {
      "id": "enrichment_data",
      "path": "${@currentAlert}.enrichmentData"
    }
  ],
  "config": {
    "fields": [
      // Fields go here
      {
        "defaultLabel": "Email Address",
        "field": "enrichment_data.email", // This enrichment field is a string of the entire hyperlink
        "type": "hyperlink" // Specifying hyperlink makes the enrichment field string a link
      },
      {
        "defaultLabel": "Address",
        "field": "bulk_contact_data.streetAddress",
        "type": "string"
      },
      {
        "field": "enrichment_data.transaction_method", // Since field is required, specify any field with a
value here. It won't appear on the page when an image is specified
        "type": "image",
        "imgSrc": "http://fsirnd0050.na.sas.com/files/files/594fee71-5cfd-4b26-aa30-65bad2ef0ed2/content" //
One way to obtain image from SAS ecosystem. File must be shared with users in order for them to see it on the
page
      }
    ]
  }
}
```

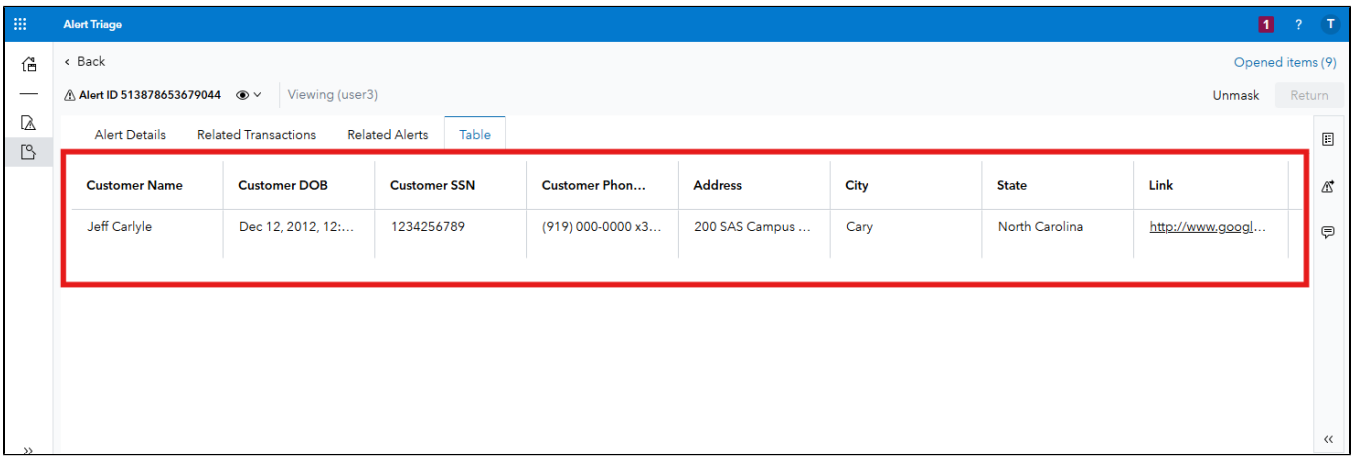


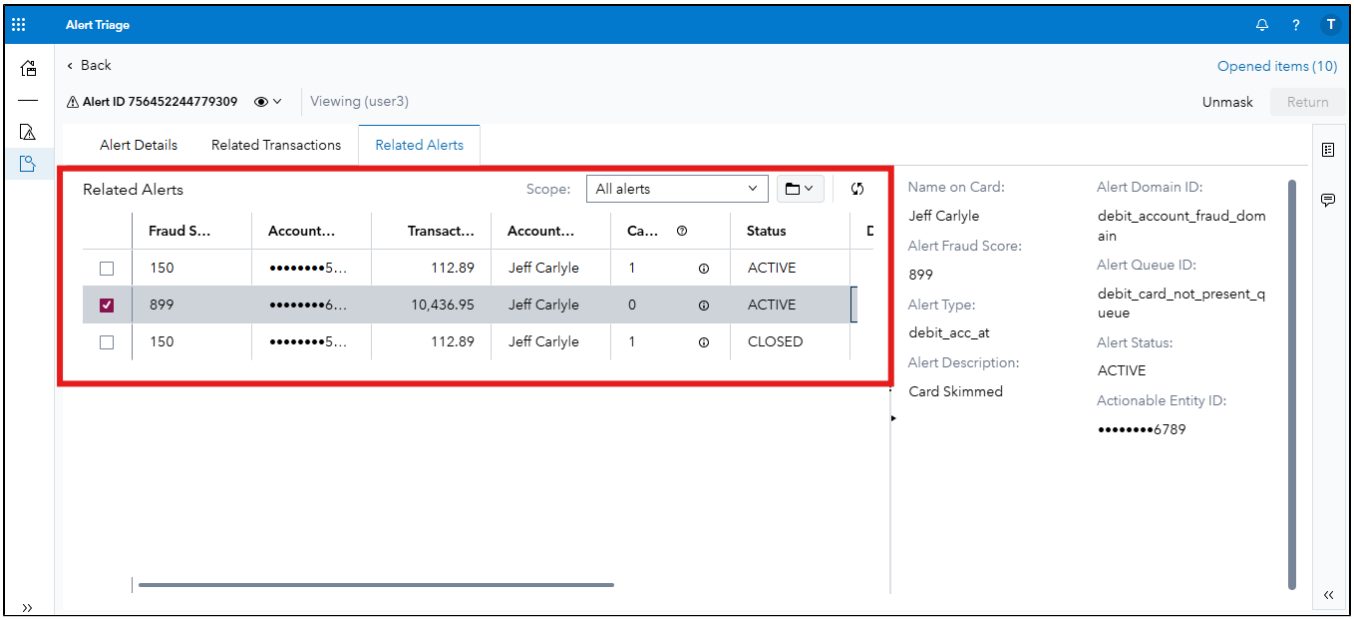
Table: Standard table, visualizes data from multiple objects

'config' Variables	Data Type	Required?	Default Value	Description

`columns`	Array of JSON objects	Yes		The columns to surface in the table
`columns[n].field`	string	Yes		The field to specify from the datasource
`columns[n].headerName`	string	No		Column label
`columns[n].hide`	boolean	No	`false`	Specifies whether column is hidden by default
`columns[n].initialSort`	string	No	`"asc"`	Specifies the initial sorting order of the column Accepted values: • `"asc"` • `"dsc"`
`columns[n].sortable`	boolean	No	`true`	Specifies if the column can be sorted by the end user
`columns[n].type`	string	No		Specifies the type of data being surfaced in the column Accepted values: • `"string"` • `"boolean"` • `"number"` • `"date"` • `"datetime"` • `"hyperlink"`
`columns[n].href`	string	Yes if columns[n].type is "hyperlink"		
`rowDensity`	string	No	`"medium"`	Specifies how closely packed the rows in the table are. "low" density = fewer rows in a given space, "high" density = more rows in a given space. Accepted values: • `"low"` • `"medium"` • `"high"`
`defaultMsgSingleSelection`	string	No	`"Make a single selection to fetch data for this component."`	Message that surfaces when incorrect number of rows are selected for singleSelection action
`useServerSideDatasource`	boolean	No	`false`	If true, this flag indicates the request be mapped to an AGGrid datasource and let it handle fetching data for the table. If false, the Table component will fetch data for the underlying AGGrid and use the AGGrid client-side row model.

Table Example JSON

```
{
  "id": "example_table",
  "type": "Table",
  "datasources": [
    // Datasources go here
    {
      "id": "contact_data",
      "externalUri": "/contact/data/${contactEntityID}"
    }
  ],
  "config": {
    "columns": [
      // Fields go here
      {
        "headerName": "Customer Name",
        "field": "contact_data.name",
        "hide": false,
        "initialSort": "dsc",
        "sortable": true,
        "type": "string"
      },
      {
        "headerName": "Customer DOB",
        "field": "contact_data.dob",
        "hide": false,
        "sortable": false,
        "type": "datetime"
      },
      {
        "field": "contact_data.state", // Since 'field' is required, put the value of a valid field here. This
will not show in the table for hyperlinks
        "headerName": "Link",
        "hide": false,
        "sortable": false,
        "href": "http://www.google.com",
        "type": "hyperlink"
      }
    ],
    "rowDensity": "low"
  }
}
```



RelatedAlerts: surfaces alert information in a specified grid definition

Variable	Data Type	Required?	Description
`configID`	string	Yes	The "gridTypeID" of the desired grid definition

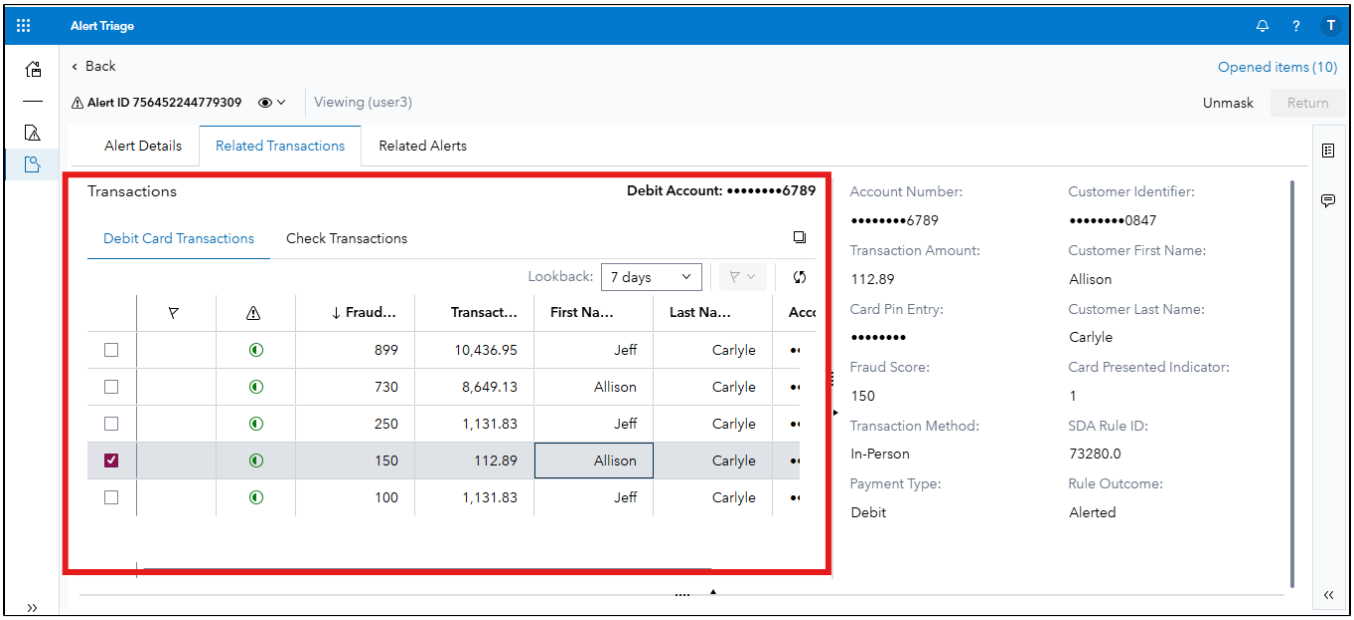
Below is an example Grid Definition JSON followed by an example RelatedAlerts component JSON:

Grid Definition Example JSON

```
{
  "id": "example_grid_definition",
  "definitionData": [
    {
      "field": "id",
      "headerNameKey": "alertsTable.columns.id.header.label",
      "hide": true
    },
    {
      "field": "enrichmentData.fraud_score",
      "filter": false,
      "headerName": "Fraud Score",
      "hide": false,
      "sortable": true,
      "type": ["enrichment", "number"]
    },
    {
      "field": "enrichmentData.transaction_amount",
      "filter": false,
      "headerName": "Transaction Amount",
      "hide": false,
      "sortable": true,
      "type": ["enrichment", "double"]
    },
    {
      "field": "actionableEntityLabel",
      "headerName": "Account Holder",
      "sortable": false
    },
    {
      "field": "alertTriggerText",
      "headerNameKey": "alertsTable.columns.alertTriggerText.header.label",
      "hide": true
    }
  ],
   // This will be the configID for the RelatedAlerts component
  "gridTypeID": "debit_card_fraud_triage_type",
  "version": 1,
  "optLockVersion": -1
}
```

RelatedAlerts Example JSON

```
{
  "id": "debit_related_alerts_pane",
  "type": "Pane",
  "config": {
    "defaultLabel": "Related Alerts"
  },
  "childComponents": [
    {
      "type": "RelatedAlerts",
      // The grid definition id. No id is needed for the RelatedAlerts component itself since the configID is
required
      "configID": "debit_card_fraud_triage_type"
    }
  ]
}
```



Transactions: surfaces transactions in a specified Transaction Viewer definition

Variable	Data Type	Required?	Description
`configID`	string	Yes	The id of the desired Transaction Viewer definition

Below, you'll find an example of a Transaction Viewer JSON. To learn more about configuring Transaction Viewers and Transaction grids, refer to this video

Example Transaction Viewer JSON

```
{
  // id to be used in the configID of the Transactions component
  "id": "debit_account_transaction_viewer",
  "indexableVariablePath": "message.debitaccount.number",
  "alertFieldValuePath": "actionableEntityID",
  "transactionGridDefinitionConfigs": [
    {
      "id": "check_transaction_grid",
      "displayOrderVal": 1
    },
    {
      "id": "debit_card_transaction_grid",
      "displayOrderVal": 0
    }
  ],
  "transactionMarkConfigs": [
    { "id": "review-required" },
    { "id": "review-in-progress" },
    { "id": "reviewed-inconclusive" },
    { "id": "declined" },
    { "id": "authorized" }
  ],
  "optLockVersion": -1
}
```

Transactions Example JSON

```
{
  "id": "debit_transactions_pane",
  "type": "Pane",
  "config": {
    "defaultLabel": "Transactions"
  },
  "childComponents": [
    {
      "type": "Transactions",
      // The id of the Traansaction Viewer. No id is needed for the Transactions component since the configID
is required
      "configID": "debit_account_transaction_viewer"
    }
  ]
}
```

Reusable Components

As mentioned before, components do not have to be defined in the page definition itself. If you want to clean up your page definition JSON, or reuse the same component configuration, **reusable components** are the perfect solution!

Reusable components are sent to the "/alerts/componentDefinitions" endpoint and are defined the same way components are defined within the "definitionData" object in the page definition. To reference a reusable component in the page, simply define the reusable component's 'id' as the 'configID' in the page definition.

Reusable Component JSON Skeleton

```
{
  "post": "/alerts/componentDefinitions",
  "contentType": "application/vnd.sas.collection.triage.component.definition+json",
  "acceptType": "application/json",
  "body": [
    [
      {
        // Reusable component's ID
        "id": "contact_data_reusable_component",
        "definitionData": {
          "type": "Pane",
          "config": { "defaultLabel": "Account Holder", "border": false },
          "childComponents": [
            {
              // Child components would go here
            }
          ],
          "version": 2
        },
        "optLockVersion": -1
      }
    ]
  ]
}
```

Page Definition Referencing Reusable Component

```
{
  "post": "/alerts/pageDefinitions",
  "contentType": "application/vnd.sas.collection.triage.page.definition+json",
  "acceptType": "application/vnd.sas.collection+json",
  "body": [
    [
      {
        "id": "page_definition",
        "optLockVersion": -1,
        "entityTypeID": "example_entity_type",
        "triageTypeID": "examply_triage_type",
        "definitionData": {
          "id": "tab_bar",
          "type": "TabBar",
          "config": { "compact": false },
          "childComponents": [
            {
              "id": "contact_details_tab",
              "type": "Tab",
              "config": {
                "defaultLabel": "Contact Details",
                "showToolbar": true,
                "flow": "horizontal"
              },
              "childComponents": [
                {
                  // Reusable component id here
                  "configID": "contact_data_reusable_component"
                }
              ]
            }
          ],
          "version": 2
        }
      }
    ]
  ]
}
```

Additionally, if there are certain 'config' fields you wish to change within the reusable component, you can modify them in the page definition!

Modified Reusable Component in Page Example

```
{
  "post": "/alerts/pageDefinitions",
  "contentType": "application/vnd.sas.collection.triage.page.definition+json",
  "acceptType": "application/vnd.sas.collection+json",
  "body": [
    [
      {
        "id": "page_definition",
        "optLockVersion": -1,
        "entityTypeID": "example_entity_type",
        "triageTypeID": "examply_triage_type",
        "definitionData": {
          "id": "tab_bar",
          "type": "TabBar",
          "config": { "compact": false },
          "childComponents": [
            {
              "id": "contact_details_tab",
              "type": "Tab",
              "config": {
                "defaultLabel": "Contact Details",
                "showToolbar": true,
                "flow": "horizontal"
              },
              "childComponents": [
                {
                  // Reusable component id here
                  "configID": "contact_data_reusable_component",
                              // You can modify reusable components in the page definition!
                                  "config": {
                                    "border": false,
                    "collapsible": true,
                                "spacing": "medium"
                                  }
                }
              ]
            }
          ],
          "version": 2
        }
      }
    ]
  ]
}
```

Reusable component modifications use a shallow merge, which means only the top-level settings are combined. If there are nested child components, objects, arrays, etc., they won't be merged together—instead, they'll be replaced entirely by the new version. See the example below for more details.

CAVEAT- Modified Reusable Component in Page Example

```
{
  "post": "/alerts/pageDefinitions",
  "contentType": "application/vnd.sas.collection.triage.page.definition+json",
  "acceptType": "application/vnd.sas.collection+json",
  "body": [
    [
      {
        "id": "page_definition",
        "optLockVersion": -1,
        "entityTypeID": "example_entity_type",
        "triageTypeID": "examply_triage_type",
        "definitionData": {
          "id": "tab_bar",
          "type": "TabBar",
          "config": { "compact": false },
```

```
          "childComponents": [
            {
              "id": "contact_details_tab",
              "type": "Tab",
              "config": {
                "defaultLabel": "Contact Details",
                "showToolbar": true,
                "flow": "horizontal"
              },
              "childComponents": [
                {
                  // Reusable component id
                  "configID": "contact_data_reusable_component",
                  // Changes to the reusable component
                  "config": {
                    "border": false,
                    "collapsible": true,
                    "spacing": "medium"
                  },
                  "childComponents": [
                    {
                        // Another reusable component id
                      "configID": "contact_information_equal_columns_reusable_component",
                      "datasources": [
                        {
                          "id": "enrichment_data",
                          "path": "${@currentAlert}.enrichmentData"
                        }
                      ],
                      // These 'config' modifications will take precedence over the reusable components 'config'
                      "config": {
                        "fields": [
                          // This field will be the only one in this instance of the reusable component. It
will replace all other fields of the reusable component rather than merging with them.
                          // If you want to add a new field to the existing reusable component, you must
redefine all its current fields in the page definition, including the new field you wish to add.
                          {
                            "field": "enrichment_data.email",
                            "type": "string",
                            "defaultLabel": "Contact email"
                          }
                        ],
                        "numCols": 3
                      }
                    }
                  ]
                }
              ]
            }
          ],
          "version": 2
        }
      }
    ]
  ]
}
```

Example Configuration Files

Below are example configuration files for both a Page Definition and a Reusable Component. These files are meant to provide a more holistic example of how you can create a page in SAS Alert Triage. Please note that, in order to implement your own page, the IDs (e.g. entityTypeID, triageTypeID, etc.) will need to match those in your SAS Alert Triage configuration.

Example Page Definition JSON

```
{
  "post": "/alerts/pageDefinitions",
  "contentType": "application/vnd.sas.collection.triage.page.definition+json",
  "acceptType": "application/vnd.sas.collection+json",
  "body": [
    [
      {
        "id": "debit_page_definition",
        "optLockVersion": -1,
        "entityTypeID": "debit_account_entity_type",
        "triageTypeID": "debit_card_fraud_triage_type",
        "definitionData": {
          "id": "debit_tab_bar",
          "type": "TabBar",
          "config": { "compact": false },
          "childComponents": [
            {
              "id": "current_alert_tab",
              "type": "Tab",
              "config": {
                "defaultLabel": "Alert Details",
                "showToolbar": true,
                "flow": "horizontal"
              },
              "childComponents": [
                {
                  "id": "alert_details_splitter",
                  "type": "Splitter",
                  "config": { "splitbarType": "verticalCollapseRight", "defaultSizeSection": "5%"},
                  "childComponents": [
                    {
                      "id": "alert_details_pane",
                      "type": "Pane",
                      "coordinates": [0, 0],
                      "config": {
                        "defaultLabel": "Alert Details",
                        "border": false
                      },
                      "childComponents": [
                        {
                          "id": "current_alert_info",
                          "type": "EqualColumns",
                          "defaultLabel": "Current Alert Information",
                          "datasources": [
                            {
                              "id": "alert",
                              "path": "${@currentAlert}"
                            },
                            {
                              "id": "enrichment_data",
                              "path": "${@currentAlert}.enrichmentData"
                            }
                          ],
                          "config": {
                            "fields": [
                              {
                                "defaultLabel": "Alert Fraud Score",
                                "field": "alert.score",
                                "type": "number"
                              },
                              {
                                "defaultLabel": "Alert Type",
                                "field": "alert.alertType",
                                "type": "string"
                              },
                              {
                                "defaultLabel": "Alert Description",
                                "field": "alert.alertTriggerText",
                                "type": "string"
                              },
                              {
```

```
                    "defaultLabel": "Alert Domain ID",
                    "field": "alert.domainID",
                    "type": "string"
                  },
                  {
                    "defaultLabel": "Alert Queue ID",
                    "field": "alert.queueID",
                    "type": "string"
                  },
                  {
                    "defaultLabel": "Alert Status",
                    "field": "alert.status",
                    "type": "string"
                  },
                  {
                    "defaultLabel": "Actionable Entity ID",
                    "field": "alert.actionableEntityID",
                    "type": "string"
                  }
                ]
              }
            }
          ]
        },
        {
          "configID": "contact_data_reusable_component",
          "coordinates": [0, 1]
        }
      ]
    }
  ]
},
{
  "id": "transacion_tab",
  "type": "Tab",
  "config": {
    "defaultLabel": "Related Transactions",
    "showToolbar": false,
    "flow": "horizontal"
  },
  "childComponents": [
    {
      "id": "page_splitter",
      "type": "Splitter",
      "config": { "splitbarType": "horizontalCollapseDown" },
      "childComponents": [
        {
          "configID": "contact_data_reusable_component",
          "coordinates": [1, 0]
        },
        {
          "id": "transaction_splitter",
          "type": "Splitter",
          "config": { "splitbarType": "verticalCollapseRight" },
          "childComponents": [
            {
              "type": "Transactions",
              "configID": "debit_account_transaction_viewer",
              "coordinates": [0, 0]
            },
            {
              "id": "transaction_action_component",
              "type": "EqualColumns",
              "coordinates": [0, 1],
              "datasources": [
                {
                  "id": "transaction_details",
                  "request": {
                    "headers": {
                      "Accept": "application/vnd.sas.triage.transaction+json",
                      "Content-Type": "application/vnd.sas.triage.transaction.details.query+json"
```

```
                                },
                                "uri": "/alertsTransaction/transactions/${@component.
debit_account_transaction_viewer.singleSelection.transactionID}?transactionTypeID=${@component.
debit_account_transaction_viewer.singleSelection.transactionTypeID}",
                                "verb": "GET"
                              }
                            }
                          ],
                          "config": {
                            "fields": [
                              {
                                "defaultLabel": "Account Number",
                                "field": "transaction_details.transactionMessage.message.debitaccount.number",
                                "type": "string"
                              },
                              {
                                "defaultLabel": "Transaction Amount",
                                "field": "transaction_details.transactionMessage.message.cardfinancial.amount",
                                "type": "double"
                              },
                              {
                                "defaultLabel": "Card Pin Entry",
                                "field": "transaction_details.transactionMessage.message.debitVariables.
cardPinEntry",
                                "type": "string"
                              },
                              {
                                "defaultLabel": "Fraud Score",
                                "field": "transaction_details.transactionMessage.message.debitVariables.
fraudScore",
                                "type": "string"
                              },
                              {
                                "defaultLabel": "Transaction Method",
                                "field": "transaction_details.transactionMessage.message.debitVariables.
transactionMethod",
                                "type": "string"
                              },
                              {
                                "defaultLabel": "Payment Type",
                                "field": "transaction_details.transactionMessage.message.debitVariables.
paymentType",
                                "type": "string"
                              },
                              {
                                "defaultLabel": "Customer Identifier",
                                "field": "transaction_details.transactionMessage.message.customer.identifier",
                                "type": "string"
                              },
                              {
                                "defaultLabel": "Customer First Name",
                                "field": "transaction_details.transactionMessage.message.customer.name",
                                "type": "string"
                              },
                              {
                                "defaultLabel": "Customer Last Name",
                                "field": "transaction_details.transactionMessage.message.customer.surname",
                                "type": "string"
                              },
                              {
                                "defaultLabel": "Card Presented Indicator",
                                "field": "transaction_details.transactionMessage.message.cardfinancial.
cardPresentedInd",
                                "type": "string",
                                "referenceListID": "card-present-indicator"
                              },
                              {
                                "defaultLabel": "SDA Rule ID",
                                "field": "transaction_details.transactionMessage.message.sas.rre.rules[0].
refId",
                                "type": "string"
```

```
                              },
                              {
                                "defaultLabel": "Rule Outcome",
                                "field": "transaction_details.transactionMessage.message.sas.decision.
outcomeName",
                                "type": "string"
                              }
                            ],
                            "numCols": 2,
                            "defaultMsgSingleSelection": "Please select a transaction to view its details here"
                          }
                        }
                      ]
                    }
                  ]
                }
              ]
            },
            {
              "id": "debit_related_alerts_tab",
              "type": "Tab",
              "config": {
                "defaultLabel": "Related Alerts",
                "showToolbar": false,
                "flow": "horizontal"
              },
              "childComponents": [
                {
                  "id": "alerts_splitter",
                  "type": "Splitter",
                  "config": { "splitbarType": "verticalCollapseRight" },
                  "childComponents": [
                    {
                      "type": "RelatedAlerts",
                      "configID": "debit_card_fraud_triage_type",
                      "coordinates": [0, 0]
                    },
                    {
                      "id": "alerts_action_component",
                      "type": "EqualColumns",
                      "coordinates": [0, 1],
                      "datasources": [
                        {
                          "id": "alert_details",
                          "request": {
                            "headers": {
                              "Accept": "application/vnd.sas.triage.alert+json",
                              "Content-Type": "application/vnd.sas.triage.alert+json"
                            },
                            "uri": "/alerts/alerts/${@component.debit_card_fraud_triage_type.singleSelection.
id}",
                            "verb": "GET"
                          }
                        }
                      ],
                      "config": {
                        "fields": [
                          {
                            "defaultLabel": "Name on Card",
                            "field": "alert_details.actionableEntityLabel",
                            "type": "string"
                          },
                          {
                            "defaultLabel": "Alert Fraud Score",
                            "field": "alert_details.score",
                            "type": "number"
                          },
                          {
                            "defaultLabel": "Alert Type",
                            "field": "alert_details.alertType",
                            "type": "string"
```

```
                        },
                        {
                          "defaultLabel": "Alert Description",
                          "field": "alert_details.alertTriggerText",
                          "type": "string"
                        },
                        {
                          "defaultLabel": "Alert Domain ID",
                          "field": "alert_details.domainID",
                          "type": "string"
                        },
                        {
                          "defaultLabel": "Alert Queue ID",
                          "field": "alert_details.queueID",
                          "type": "string"
                        },
                        {
                          "defaultLabel": "Alert Status",
                          "field": "alert_details.status",
                          "type": "string"
                        },
                        {
                          "defaultLabel": "Actionable Entity ID",
                          "field": "alert_details.actionableEntityID",
                          "type": "string"
                        }
                      ],
                      "numCols": 2,
                      "defaultMsgSingleSelection": "Please select an alert to view its details here"
                    }
                  }
                ]
              }
            ]
          }
        ],
        "version": 2
      }
    }
  ]
]
}
```

Example Reusable Component JSON

```
{
  "post": "/alerts/componentDefinitions",
  "contentType": "application/vnd.sas.collection.triage.component.definition+json",
  "acceptType": "application/json",
  "body": [
    [
      {
        "id": "contact_data_reusable_component",
        "definitionData": {
          "type": "Pane",
          "config": { "defaultLabel": "Account Holder", "border": false },
          "childComponents": [
            {
              "id": "contact_information_equal_columns",
              "type": "EqualColumns",
              "datasources": [
                {
                  "id": "bulk_contact_data",
                  "externalUri": "/contact/data/${contactEntityID}"
                },
                {
                  "id": "enrichment_data",
                  "path": "${@currentAlert}.enrichmentData"
                }
```

```
              ],
              "config": {
                "fields": [
                  {
                    "defaultLabel": "First Name",
                    "field": "enrichment_data.customer_first_name",
                    "type": "string"
                  },
                  {
                    "defaultLabel": "Last Name",
                    "field": "enrichment_data.customer_last_name",
                    "type": "string"
                  },
                  {
                    "defaultLabel": "Phone Number",
                    "field": "bulk_contact_data.phone",
                    "type": "string"
                  },
                  {
                    "defaultLabel": "Email Address",
                    "field": "enrichment_data.email",
                    "type": "hyperlink"
                  },
                  {
                    "defaultLabel": "Address",
                    "field": "bulk_contact_data.streetAddress",
                    "type": "string"
                  },
                  {
                    "defaultLabel": "City",
                    "field": "bulk_contact_data.city",
                    "type": "string"
                  },
                  {
                    "defaultLabel": "State",
                    "field": "bulk_contact_data.state",
                    "type": "string"
                  },
                  {
                    "field": "enrichment_data.transaction_method",
                    "type": "image",
                    "imgSrc": "http://fsirnd0050.na.sas.com/files/files/594fee71-5cfd-4b26-aa30-65bad2ef0ed2
/content",
                    "defaultLabel": "Image"
                  }
                ]
              }
            }
          ],
          "version": 2
        },
        "optLockVersion": -1
      }
    ]
  ]
}
```